

Rational Design of UvsX Recombinase Variants for Enhanced Performance in Recombinase Polymerase Amplification Applications

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Cite This: *Biochemistry* 2025, 64, 2025–2038



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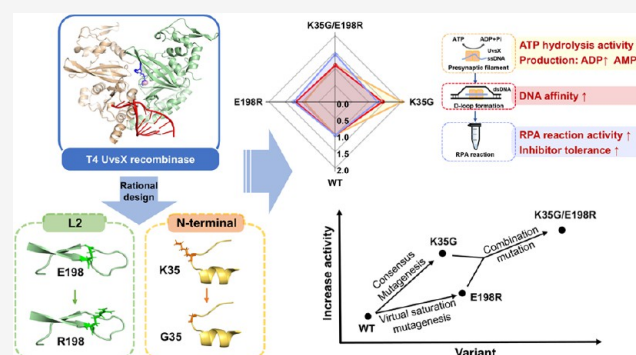
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ABSTRACT: Homologous recombination is a vital biological process for DNA repair, genomic stability, and genetic diversity, driven by the RecA/Rad51 recombinase family. However, as a T4 bacteriophage recombinase homologous to RecA/Rad51, UvsX has limited *in vitro* performance during recombinase polymerase amplification (RPA) due to ATP utilization and DNA affinity. In this study, UvsX was rationally engineered to enhance these properties through homology modeling, virtual saturation mutations, and consensus mutation strategies. Targeted mutagenesis produced UvsX variants (E198N, E198R, E198K, and K35G) with a $16 \pm 4\%$ to $39 \pm 6\%$ improvement in RPA activity, while the double mutant K35G/E198R showed an increase of up to $43 \pm 4\%$. Structural analysis revealed that the K35G/E198R mutation enlarged ATP-binding pockets and increased the positive surface potential of DNA-binding sites, resulting in a $12 \pm 4\%$ improvement in ATP utilization and more ADP and less AMP generated, a $10 \pm 2\%$ enhancement in DNA interaction compared to the wild-type, and better inhibitor tolerance. These findings establish a foundation for the rational optimization of recombinases in nucleic acid amplification and promote their potential for industrial RPA applications.



1. INTRODUCTION

Homologous recombination (HR) is a fundamental biological process that is crucial in maintaining genomic stability, repairing DNA double-strand breaks, promoting genetic diversity, and generating immune diversity. Beyond its intrinsic biological significance, HR has been widely applied in gene therapy and medical research.^{1–3} The RecA/Rad51 recombinase family of enzymes, central to HR, is essential for the error-free repair of double-strand DNA breaks and the restart of stalled replication forks.^{4,5} DNA recombination does not occur spontaneously, even in the presence of DNA breaks or gaps; it requires recombinase enzymes to facilitate the reaction.^{6,7} T4 UvsX protein derived from the T4 phage, as a member of the RecA/Rad51 family, is responsible for recognizing homology between homologous DNA partners and catalyzing the strand exchange,⁸ and exhibits DNA-dependent and ATPase activity in the process.⁹

Recent studies have highlighted the utility of UvsX recombinase in fields such as viral detection, food safety testing, isothermal amplification, homologous recombination cloning, and gene editing-assisted detection.^{10–13} Notably, UvsX has shown great promise in recombinase polymerase amplification (RPA), one of the most advanced isothermal amplification techniques. RPA has several advantages over traditional polymerase chain reaction (PCR), including

simplicity of operation, rapid amplification (typically 20–30 min), and high specificity and sensitivity comparable to PCR.^{14–16} During RPA, UvsX is crucial for the formation of the D-loop structure,¹⁷ which facilitates DNA strand exchange and template synthesis.¹⁸ The primary structural domain of protein comprises ATP and DNA binding sites. The ATP binding site is located in the P-loop (residues 59–67), while DNA binding primarily occurs through two loops in the core domain, L1 (residues 152–165) and L2 (residues 198–211).¹⁹ These structural features enable UvsX to participate in the strand exchange reactions central to RPA. Despite its pivotal role during *in vitro* reactions, the practical application of UvsX remains limited due to several challenges. The activity and stability of UvsX in the multienzyme reaction systems remain suboptimal. Additionally, UvsX exhibits low ATP utilization efficiency and insufficient DNA affinity, which makes it a rate-limiting factor in the RPA reaction. These limitations hinder

Received: February 21, 2025

Revised: March 22, 2025

Accepted: April 15, 2025

Published: April 22, 2025



the advancement of RPA technology and its widespread application across various industries.

In addition to its role in RPA, UvsX recombinase may have potential *in vitro* applications in other fields, such as DNA repair, homologous recombination, and genetic engineering.^{20–22} In synthetic biology, UvsX could also serve as a promising tool for creating new recombination-based systems or improving existing gene editing and recombination-mediated DNA manipulation systems. It may also have future applications in industrial biotechnology to optimize gene expression.^{23,24} However, the low DNA-binding affinity and stability of UvsX reduces the efficiency of the DNA repair process, especially when efficient strand invasion and homology search are required. Therefore, there is an urgent need to modify the UvsX to improve its affinity for DNA.

In this study, we reconstructed the structure and DNA binding pocket of UvsX with a rational design strategy. We first performed *in silico* scanning on key residues within 3 Å of the binding site and then utilized virtual saturation mutagenesis and consensus mutagenesis to modify the core structural domains of UvsX. Four mutants (E198N, E198R, E198K, and K35G) were identified, which all showed increased RPA activity compared to the wild type (WT). The double mutant K35G/E198R, which exhibited the highest RPA activity, was then selected for further experimental verifications and mechanism analysis. This work provides valuable insights into the design of UvsX mutants with enhanced activity, which may facilitate the development of more efficient isothermal amplification technologies and improve the understanding of recombinase-mediated DNA strand exchange mechanisms.

2. MATERIALS AND METHODS

2.1. Construction of Recombinant Plasmids. The T4 UvsX gene (GenBank ID: KJ477686.1) was synthesized by GENEWIZ Biotechnology Co. Ltd. (Suzhou, China), and cloned into the NdeI and XhoI restriction sites of the pET-28a (+) plasmid to construct the pET-UvsX recombinant plasmid. To address potential design flaws in the pET series that may limit protein yield, the translation initiation region (TIR) of the vector T7 promoter was modified and optimized.²⁵ The recombinant construct included 6× His tags and thrombin cleavage sites at both the N- and C-terminals to facilitate protein purification. Primer sequences used in this study are listed in Table S1.

2.2. Strain Culture and Protein Purification. The *Escherichia coli* BL21(DE3) carrying pET-UvsX (PDB: 3IO5) was cultured in LB medium with 0.1 mM kanamycin at 37 °C and 220 rpm. Protein expression was induced with 0.05 mM IPTG at an OD of ~0.8, followed by incubation at 16 °C and 160 rpm for 20 h.

Cells were harvested by centrifugation at 8000g for 5 min. The pellet was resuspended in lysis buffer (50 mM phosphate, pH 7.2, 500 mM NaCl, 0.1 mM PMSF). After mechanical disruption, the lysate was centrifuged at 10,000g for 1 h. The supernatant was subjected to affinity chromatography using Ni-NTA resin. The eluted protein was dialyzed against storage buffer (20 mM Tris-HCl, pH 8.0, 300 mM NaCl, 1 mM dithiothreitol (DTT), 0.1 mM EDTA, 20% v/v glycerol) and concentrated using a 10 kDa ultrafiltration device. Protein purity and molecular weight were assessed by 12% sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE), and the protein concentration was determined using a BCA Protein Assay Kit (Solarbio, Beijing). A fast mutagenesis

system (GENEWIZ; Suzhou, China) was used to execute site-directed mutagenesis. The primers used for cloning and site-directed mutagenesis are described in Table S2.

2.3. Mutation Site Selection Strategy. A structural model of the UvsX hexamer bound to the oligonucleotide dT₇ was constructed using AlphaFold 3.²⁶ The Molecular Mechanics/Generalized Born Surface Area (MM/GBSA) method was used to estimate energy changes, calculating the free energy differences before and after mutagenesis to evaluate the effect of structural modifications on protein stability.²⁷ Alanine scanning and virtual saturation mutagenesis of key residues within 3 Å of the dT₇ binding site were performed using FoldX.²⁸ Mutation-based energy screening was then employed to modify the core structural domain of UvsX.

The Consensus Finder online tool (<http://kazlab.umn.edu/>) was used to calculate the frequency of occurrence of each amino acid in homologous sequences of UvsX. A conservative amino acid threshold of greater than 50% was set, and a consensus mutagenesis strategy was employed to mutate nonconservative amino acids to conserved ones.

2.4. Melting Temperature (*T_m*) Measurement. Differential scanning fluorimetry (DSF) is a widely used technique to evaluate the thermal stability of proteins.²⁹ The assay was performed in triplicate using a fluorescence quantitative PCR instrument (ABI QuantStudio 5, Applied Biosystems) with a 96-well plate format. Each reaction contained 60 mM Tris-HCl (pH 8.0), 80 mM CH₃COOK, 4 mM DTT, 14 mM MgCl₂, 4 μg of protein, and 1× SYPRO orange dye. Fluorescence intensity was recorded at an excitation wavelength of 465 nm and an emission wavelength of 580 nm, with the temperature ramped from 25 to 100 °C at 0.05 °C·s^{−1} and measurements taken every 2 s. The protein melting temperature (*T_m*) was derived from the first derivative of the melting curve.³⁰

2.5. RPA Reaction. The RPA detection system was used to amplify sequences from the pUC19 plasmid, with the amplified nucleotide sequences and corresponding primer sequences provided in Figure S1. The total reaction volume was 50 μL, with the reaction conditions as follows: 120 ng·μL^{−1} of UvsX, 20 ng·μL^{−1} of UvsY, 30 ng·μL^{−1} of Bsu DNA polymerase, 200 ng·μL^{−1} of gp32, 100 ng·μL^{−1} of creatine kinase, 30 mM Tris-HCl buffer (pH 8.0), 40 mM CH₃COOK, 2 mM DTT, 20 mM phosphocreatine, 7.5% (w/v) PEG 35,000, 2.5 mM ATP, 0.45 mM dNTPs, 14 mM MgCl₂, 0.3 μM lacZα-2F primer, 0.3 μM lacZα-2R primer, and 10⁶ copies of pUC19 plasmid. The reaction mixture was incubated at 41 °C for 30 min. Following incubation, an equal volume of DNA extraction phenol (Solarbio, Beijing) was added for nucleic acid extraction. The mixture was centrifuged at 12,000g for 5 min, and the supernatant was collected. The extracted products were separated on a 2.0% (w/v) agarose gel. The percentage increase in activity was calculated based on the band intensity from agarose gel electrophoresis. Gel images were analyzed using ImageJ software to measure the average gray values of the target bands, which were then normalized to WT as an internal ref³¹. The percentage increase in activity of mutants relative to WT was calculated using the equation. All experiments were performed in triplicate:

$$\text{percentage increase in activity (\%)} = \left(\frac{\text{normalized mutant activity}}{\text{normalized WT activity}} - 1 \right) \times 100 \quad (1)$$

The real-time fluorescence RPA reaction included 1× SYBR Green I dye in addition to the standard RPA components. Fluorescence intensity was recorded using the ABI QuantStudio 5 real-time fluorescence quantitative PCR system. The samples were incubated at 41 °C for 30 min, and the fluorescence signal was measured every 10 s over 180 cycles using the preset SYBR detection channel in QuantStudio 5 software. All experiments were performed in triplicate.

2.6. Determination of ATPase Activity. In this study, the inorganic phosphate (PO_4^{3-}) generated during ATP hydrolysis was measured using the malachite green-molybdenum heteropolyacid spectrophotometric method to evaluate ATPase activity.³² Total phosphate content was determined using a standard curve generated with phosphate standards at concentrations of 0, 0.05, 0.1, 0.15, 0.2, and 0.25 $\mu\text{g}\cdot\text{mL}^{-1}$. For standard curve preparation, 80 μL of the colorant (ammonium molybdate/concentrated sulfuric acid/malachite green at a ratio of 40/36/34) and 40 μL of 1% (w/v) polyethylene glycol solution (PVA) were added to each reaction mixture, with the final volume adjusted to 1 mL using distilled water. After thorough mixing, the reaction was incubated at 37 °C for 20 min, and absorbance was measured at 620 nm (Eppendorf BioPhotometer D30, Germany).

To evaluate UvsX ATPase activity, reaction mixtures (50 μL) were prepared containing 3 μg of UvsX, 0.3 mM ATP, 25 μL of buffer (60 mM Tris-HCl, pH 8.0, 80 mM CH_3COOK , 4 mM DTT). The reactions were incubated at 37 °C for 10 min, then immediately cooled on ice to terminate the reaction. Color reagent and 1% (w/v) PVA solution were subsequently added to adjust the final volume to 1 mL. After thorough mixing, the reaction was incubated at 37 °C for 20 min, and absorbance at 620 nm was recorded. All experiments were performed in triplicate. To evaluate the reaction kinetics of ATP hydrolysis by the Michaelis–Menten equation, the equation is as follows:

$$v = \frac{V_{\max} \times S}{K_m + S} \quad (2)$$

Here, $V_{\max}$ is the reaction rate of UvsX when saturated with ATP, S represents the concentration of ATP, and K_m is the Michaelis–Menten constant.

2.7. Determination of Circular Dichroism. The secondary structure of the protein was analyzed in the far-UV region using a circular dichroism (CD) spectrophotometer (JASCO J-1700, Japan) under a nitrogen atmosphere at 25 °C. The purified protein was diluted to 0.05 $\mu\text{g}\cdot\mu\text{L}^{-1}$ in buffer (2 mM Tris-HCl, pH 7.7, 10 mM KCl, 0.1 mM EDTA) and loaded into a 1.0 mm path length quartz cuvette. Continuous spectra were recorded in the wavelength range of 180–280 nm with a data interval of 0.5 nm, a response time of 4.0 s, and a bandwidth of 1.0 nm, at a scanning speed of 100 nm·min^{−1}. Ellipticity data were collected continuously. The spectral data were smoothed using the noise-filtering function in the Spectra Analysis software.³³ The final reported spectrum was the average of three scans, with the buffer blank spectrum subtracted. The secondary structure of the protein was estimated using the CD Multivariate SSE software.^{34,35}

2.8. Determination of Substrate Binding Affinity. Protein binding activity to nucleic acids was evaluated using a fluorescence polarization (FP) test. A 50 μL reaction mixture, containing 10 mM Tris-HCl (pH 7.5), 10 mM MgCl_2 , 2.5 nM 5'-FAM labeled oligonucleotides, 1.2 mM ATP, and 7 $\mu\text{g}\cdot\mu\text{L}^{-1}$

UvsX, was incubated for 10 min at 37 °C. FP measurements were then taken using a multimode microplate reader (Tecan Spark, Switzerland). The measured anisotropy values were processed by Hill Equation and the corresponding dissociation constants K_d were calculated, respectively. The equation is as follows:

$$\Delta r = \frac{\Delta r_{\max} \times p^n}{K_d + p^n} \quad (3)$$

Here, $\Delta r_{\max}$ is the maximum change in fluorescence anisotropy, p represents the concentration of UvsX, K_d is the dissociation constant of UvsX for DNA, and n is the Hill coefficient.

2.9. HPLC Analysis. All samples were detected by high-performance liquid chromatography (HPLC) (Agilent Technologies 1200 Series) equipped with a ZORBAX Eclipse Plus C18 reversed-phase column (Agilent Technologies, Analytical 4.6 mm × 250 mm 5 μm) at 27 °C for the analysis of ATP hydrolysis products. For the mobile phase, buffer A (acetonitrile) and buffer B from ATP, ADP, and AMP Content HPLC Assay Kit (Solarbio, Beijing) were used at a flow rate of 1.0 mL·min^{−1}. Buffer B with a mobile phase of 97% (v/v) in 20 min. Hydrolysis product ATP, ADP, and AMP were monitored at 254 nm and the retention times were 13.8, 9.7, and 8.4 min, respectively.

2.10. RPA-CRISPR/Cas12a Coupled Detection of HPV16. RPA-CRISPR/Cas12a coupling was used to detect HPV16, and the HPV16-RPA system amplified nucleotide sequences and corresponding primer sequences provided in Figure S1.³⁶ The total reaction volume was 50 μL , with the reaction conditions as follows: 5 μL production with HPV16-RPA detection, 50 nM LbCas12a, 50 nM crRNA (5'-UAAUUUCUAC UAAGUGUAGA UAGGAGUACC UAC-GACAUGG GG-3'), 300 nM ssDNA (5'-FAM TTATT-BHQ1-3'), 5 μL 10× NEBuffer 2.1 (500 mM NaCl, 100 mM Tris-HCl, 100 mM MgCl_2 , 1 mg· μL^{-1} BSA, pH 7.9). The crRNA was designed using EZassay Biotechnology (<https://www.ezassay.com/en/rna>), the amplicon sequence of RPA was input and the origin of Cas12a was selected as LbCas12a, choose the most efficient crRNA use of synthesis. The reaction mixture was incubated at 37 °C for 20 min, and then the fluorescence signals were photographed and recorded by Blue light-emitting diode (LED) Transilluminator (LABGIC BL-15, China) for qualitative analysis. Real-time fluorescent CRISPR/Cas12a assays were performed by recording the fluorescence intensity using the ABI QuantStudio 5 real-time PCR system. The samples were incubated at 37 °C for 20 min, and the fluorescence signal was measured every 30 s over 40 cycles using the preset FAM detection channel in QuantStudio 5 software. All experiments were performed in triplicate.

In the inhibitor tolerance test, inhibitors were included in addition to the standard RPA components: hemoglobin, cervical mucus, and 0.6% (w/v) antifungal agent miconazole.

2.11. SPSS Analysis. Data were analyzed using variance analysis by IBM SPSS Statistics for Windows, Version 26.0 (IBM Corp., Armonk, N.Y.). A one-way ANOVA was conducted to compare the means of the dependent variable across different levels of independent variables. Assumptions of normality and homogeneity of variance were tested before analysis. The significance level was set at $p < 0.05$.

2.12. Molecular Dynamics Simulation. The dimeric models of the K35G/E198R mutant and wild-type UvsX bound to ATP, Mg^{2+} , and dT₇ were predicted using the

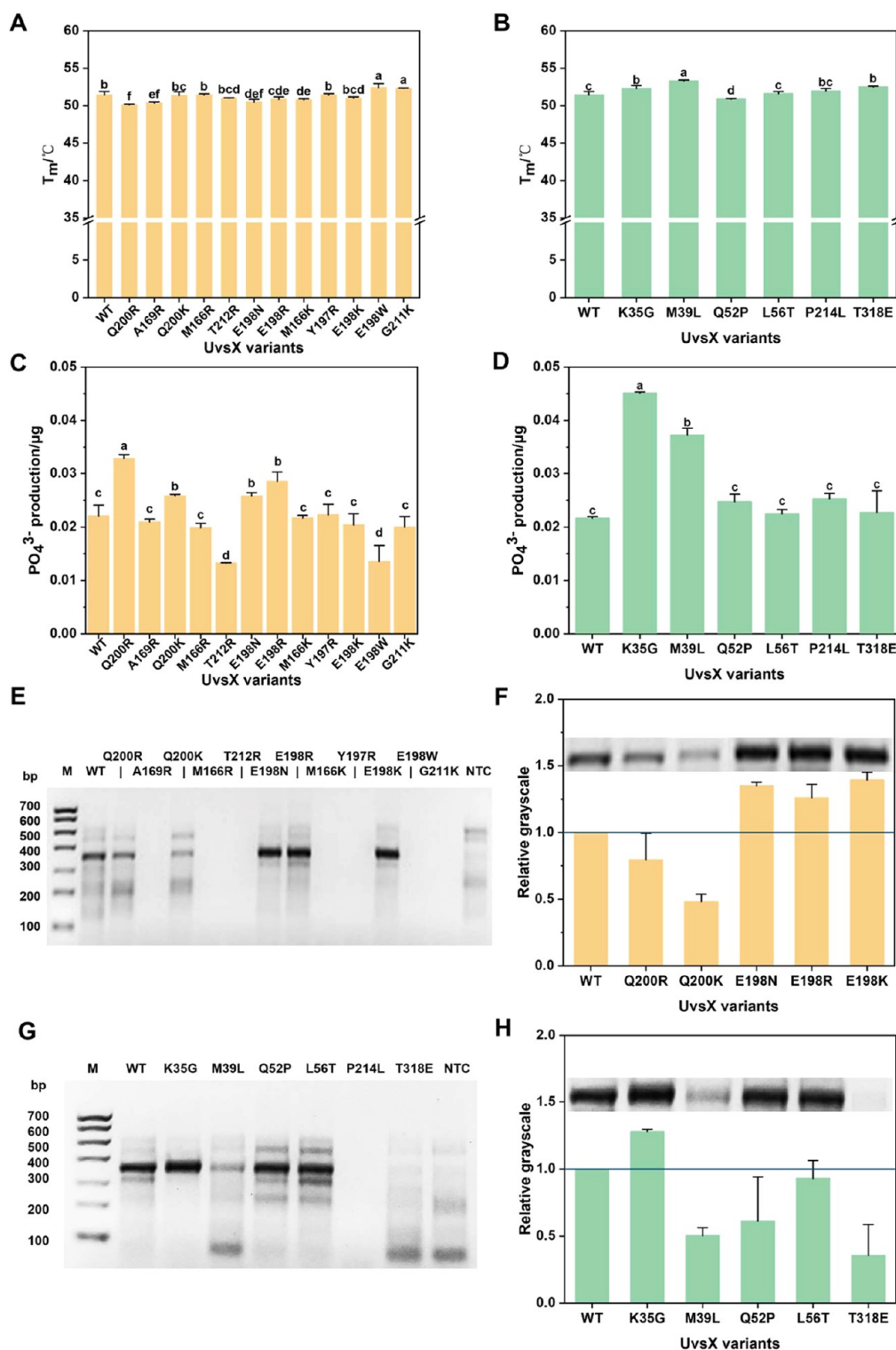


Figure 2. WT and (A) T_m values of virtual saturation mutants. The Y-axis represents the mean $\pm$ standard deviation (SD) of repeated measurements. Statistical significance was determined by one-way analysis of variance (ANOVA) followed by multiple comparisons. Different letters between bars indicate significant differences at $p < 0.05$ (e.g., "a" and "b" are significantly different), while the same letter between bars indicates no significant difference (e.g., "a" and "ab" are not significantly different). (B) T_m values of consensus mutants. (C) The ATP hydrolysis activity of the virtual saturation mutant was determined by the production of PO_4^{3-} . (D) The ATP hydrolysis activity of the consensus mutant was determined by the production of PO_4^{3-} . (E) The effects of virtual saturation mutants in RPA. (F) Relative activity of virtual saturation mutants in RPA. (G) The effects of consensus mutants in RPA. (H) Relative RPA activity of consensus mutants in RPA.

graphical representation often referred to as a dynamic cross-correlation map (DCCM) was used.⁴⁴ The model of UvsX with ADP, Mg^{2+} , and dT₇ was constructed in the same way.

Prior to the simulations, the pocket of each protein was oriented to face the Z-axis. Each resulting structure was used as the model to perform pulling dynamics simulations at a pulling

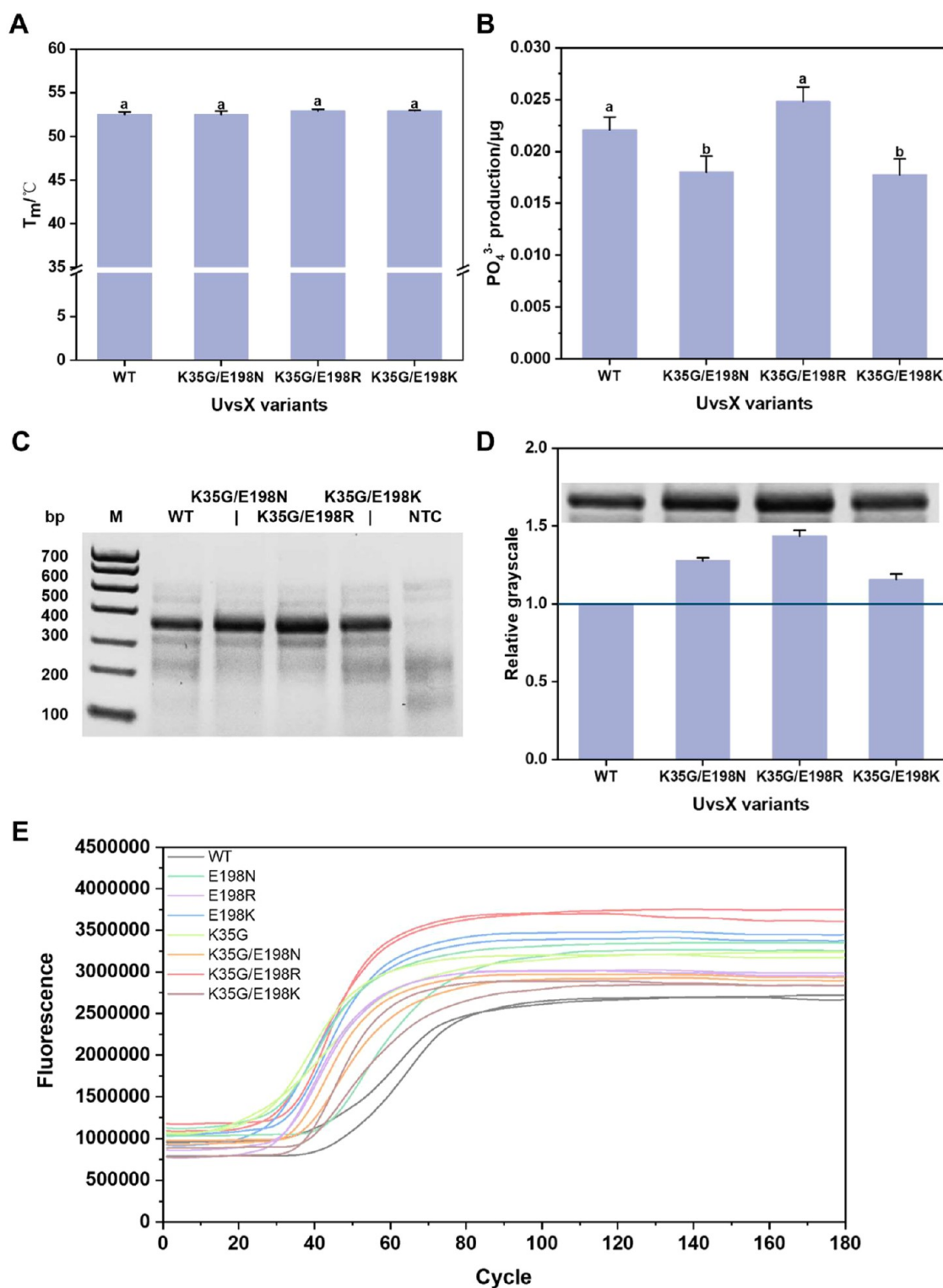


Figure 3. WT and double-point mutation of (A) T_m . The Y-axis represents the mean $\pm$ SD of replicates. Statistical significance was assessed using one-way analysis of variance (ANOVA) followed by multiple comparisons. Different letters above the bars denote statistically significant differences ($p < 0.05$; e.g., "a" and "b" indicate significant differences), whereas identical letters indicate no statistically significant differences (e.g., "a" and "ab" are not significantly different). (B) The ATP hydrolysis activity was determined by the production of PO_4^{3-} . (C) Effects in RPA. (D) Relative RPA activity. (E) Real-time RPA reaction of WT and all positive mutations, with amplification curves plotted using fluorescence signals generated by SYBR Green I dye (Y-axis) relative to time (X-axis).

speed of $0.01 \text{ nm} \cdot \text{ps}^{-1}$, with the output pulling force averaged over 1000 steps.

The UvsX-DNA complex was placed in a cubic box with a 1.0 nm margin. The simulation process was as follows: (1) a

steepest descent energy minimization was performed for 2000 steps to achieve the minimum energy configuration; (2) the system was heated to 314.15 K over 50 ps; (3) production simulations were carried out at a constant temperature of

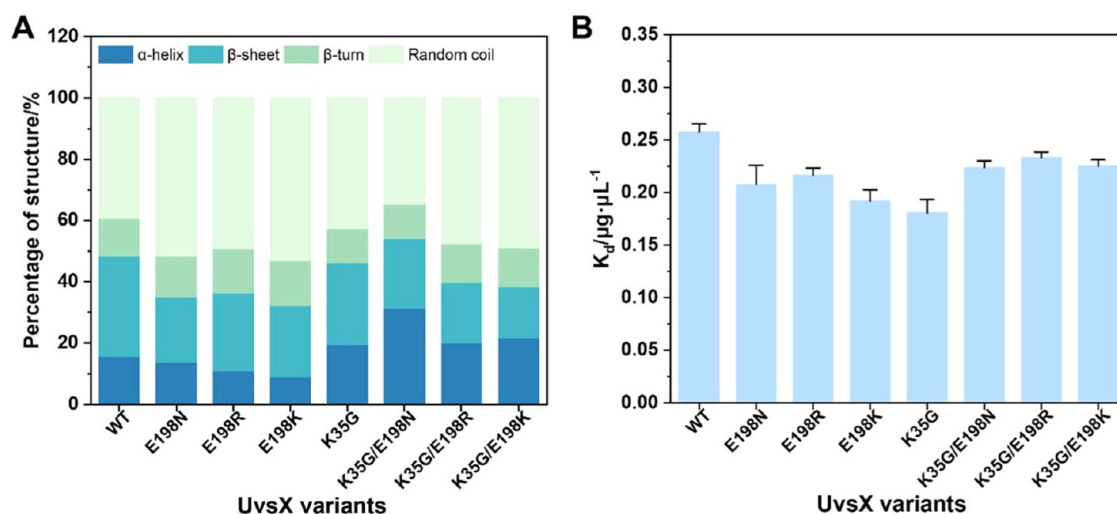


Figure 4. WT and all RPA positive mutations in (A) secondary structure composition. (B) Affinity for substrate DNA.

314.15 K using the NPT ensemble.⁴⁵ Long-range electrostatic interactions were estimated using the smooth particle mesh Ewald (PME) method.⁴⁶ Visual molecular dynamics program (VMD) and PyMOL (<https://www.pymol.org/>) were used to visualize the interactions and MD trajectories.⁴⁷ The GROMACS trajectories binding free energy of UvsX-DNA were calculated by s_mmpbsa.⁴⁸

3. RESULTS AND DISCUSSION

3.1. Identification of UvsX Directed Evolutionary Key Site. The recombinase UvsX is essential for the homologous pairing of nucleic acids, functioning by first forming presynaptic filaments with ssDNA and then facilitating homologous pairing with dsDNA. These two key steps represent the primary rate-limiting stages of the RPA reaction.¹⁷ To optimize the performance of UvsX, two rational design strategies were systematically implemented (Figure 1A,B). First, virtual saturation mutagenesis was performed to screen for candidate mutation sites with a Gibbs free energy change ($\Delta\Delta G^{\text{fold}}$) lower than $-0.5 \text{ kcal}\cdot\text{mol}^{-1}$ (Figure 1C). These candidate mutations included Q200R, A169R, Q200K, M166R, T212R, E198N, E198R, M166K, Y197R, E198K, E198W, and G211K. Second, a consensus mutagenesis approach was applied, focusing on conserved residues that are crucial for protein function. The most frequent amino acids at these sites were selected to enhance UvsX stability. Residues with nearly 100% conservation indicate their importance, and mutations in these areas are likely to disrupt function. Therefore, six mutations with over 50% conservation were identified for further testing: M39L, Q52P, L56T, K35G, P214L, and T318E (Figure 1D). This analysis helps to predict the potential structural and functional impact of the mutations.

3.2. Rational Design of UvsX to Improve *In Vitro* Reactivity. The functional and structural impacts of UvsX mutations were evaluated using three key indicators: RPA activity, ATPase activity, and thermal stability. The strand displacement activity of various mutants was assessed using the pUC19-RPA detection system. Some mutations significantly enhanced RPA activity, while others impaired it (Figure 2E,G). Notably, virtual saturation mutations E198K, E198N, and E198R increased RPA activity by $39 \pm 6\%$, $35 \pm 3\%$, and $26 \pm 5\%$, respectively, compared to WT (Figure 2F). The K35G mutation, a consensus mutation, also enhanced RPA activity by

$28 \pm 2\%$ (Figure 2H). ATPase activity, a critical measure of UvsX functionality,⁴⁹ showed variable effects among the mutants. K35G exhibited ATPase activity 2.08 times higher than WT, while T212R displayed a reduced activity of only 0.60 times that of WT (Figure 2C,D). Thermal stability analysis indicated that the T_m values of all mutants fluctuated within 2°C compared to WT, with no significant structural destabilization observed (Figure 2A,B). Among the mutants, M39L demonstrated the highest thermal stability, with a T_m of 53°C .

To further explore the positive effects, three double mutants (K35G/E198N, K35G/E198R, and K35G/E198K) were constructed. Their T_m were similar to WT (Figure 3A). Among these, K35G/E198R showed a slight increase in ATPase activity but the greatest enhancement in RPA activity, with a $43 \pm 4\%$ increase (Figure 3B–D). Real-time fluorescence RPA assays confirmed that the fluorescence intensity trends corresponded to the band intensities of amplified products observed in agarose gel electrophoresis (Figure 3E).

Conformational changes and DNA binding differences were assessed using CD and FP. The CD spectrum provided insight into secondary structural changes (Figure S3). K35G mutation increased α -helix content (Figure 4A), implying enhanced DNA binding. In contrast, Glu198 mutations (E198K, E198N, E198R) decreased both α -helix and β -sheet content, likely due to disrupted hydrogen bonding and hydrophobic interactions, resulting in lower T_m . However, DNA binding is not only affected by α -helix content but also involves various factors such as electrostatic interactions and conformational changes. The negatively charged Glu198 is located at the DNA binding site. The substitution of Glu198 with electroneutral asparagine may enhance structural flexibility and facilitate DNA sliding. And the loss of electrostatic repulsion contributes to a more stable and favorable interaction between the protein and DNA. Meanwhile, both arginine and lysine carry positive charges, promoting stronger electrostatic attraction with the negatively charged DNA backbone. And both of them have longer side chains, that can form salt bridges with the DNA backbone, which could potentially improve DNA binding and catalytic activity. Double mutations further increased α -helix content, balancing flexibility and stability, and enhancing catalytic efficiency. The dissociation constant (K_d) was used to quantify

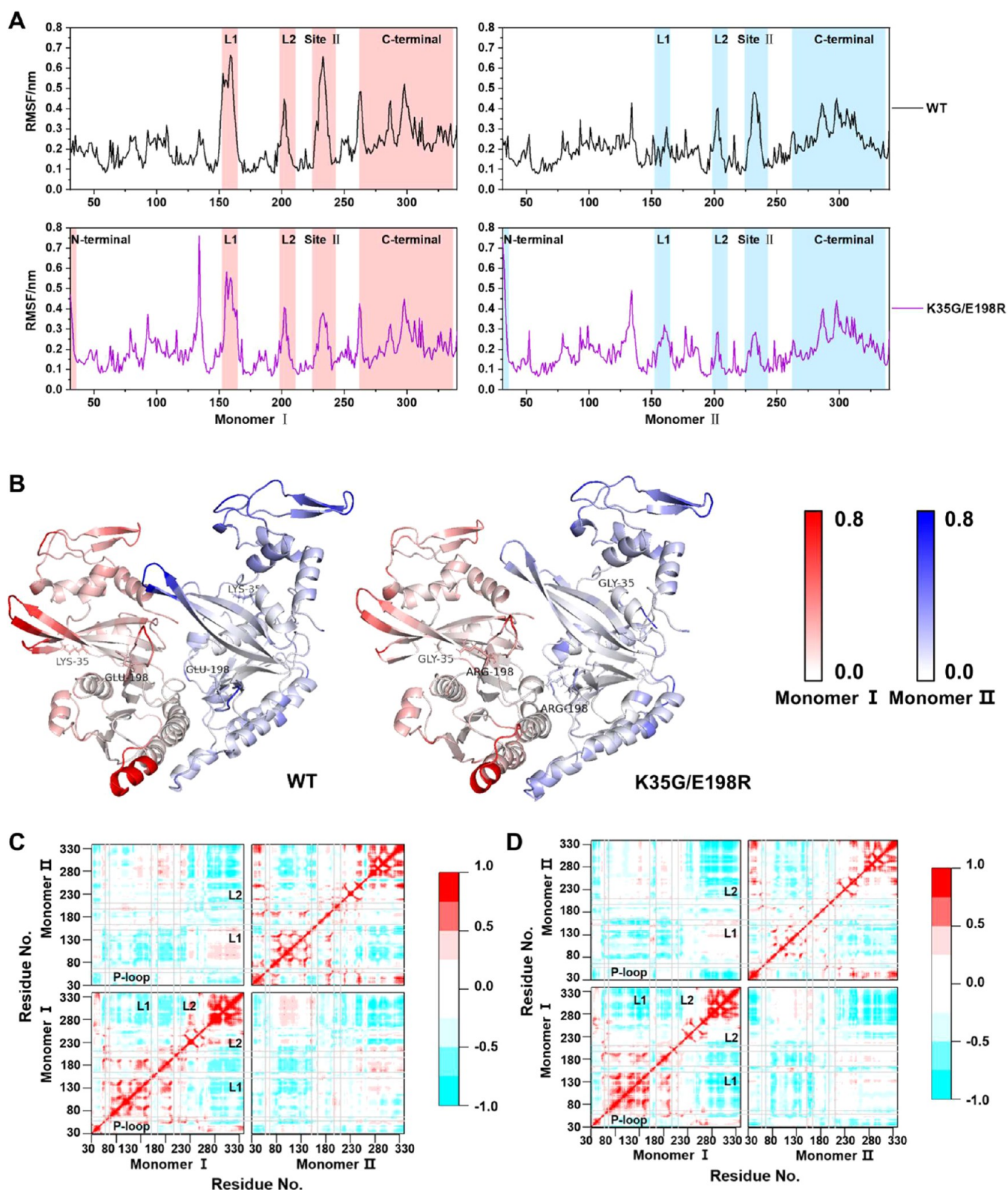


Figure 5. (A) RMSF (N-terminal, C-terminal, L1, L2 labeled in the curves). (B) Visualization of backbone flexibility for WT and K35G/E198R (for monomer I the intensity of fluctuations is indicated by a blue gradient, and for monomer II by a red gradient). Dynamic interrelationship diagram between UvsX recombinase residues: (C) WT (P-loop, L1, and L2 are marked in gray boxes, respectively; red areas indicate positive correlation, blue areas indicate negative correlation, and color density indicates the degree of correlation) and (D) K35G/E198R.

DNA binding affinity, K_d is inversely proportional to binding affinity, with lower K_d values indicating stronger DNA binding. FP showed that all positive mutants exhibited lower K_d for

ssDNA (Figure 4B, with detailed fitting in Figure S4), indicating stronger DNA binding, which is consistent with increased RPA activity.

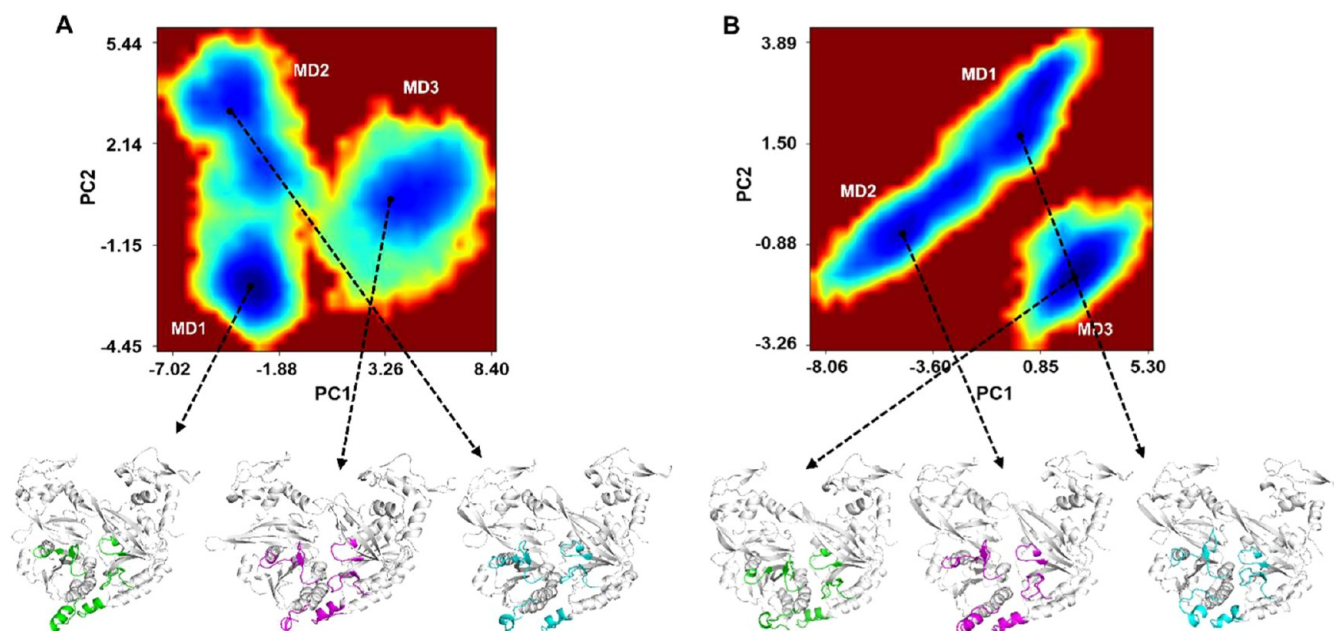


Figure 6. FELs and lowest-energy conformations of (A) WT and (B) K35G/E198R, with potential energy color-coded from red (high) to blue (low). Key conformational differences are highlighted in green (low free energy), magenta (intermediate), and blue (high free energy) for comparison.

3.3. Illustrating the Mechanism of the K35G/E198R Mutants with Increased Activity. To elucidate the mechanism underlying the activity enhancement of the mutant, we performed 100 ns MD simulations for both the WT UvsX and the K35G/E198R mutant. The Ramachandran plots (Figure S5) revealed that the majority of the residues are located within the black contour area, suggesting that both protein structures are well constructed. Analysis of the resulting simulation trajectories provided RMSD, R_g , and SASA of the proteins (Figure S6). RMSD data indicated that the system reached equilibrium within 100 ns at 314.15 K. The R_g value, which measures the structural compactness of the protein,⁴⁹ was higher for the WT than for K35G/E198R. SASA is used to evaluate the extent of the protein's surface exposure to the solvent,⁵⁰ and displayed comparable values for both proteins. These results suggested that the WT may have a more loosely conformation than the K35G/E198R mutant.

RMSF values are used to quantify the fluctuations of individuals. The RMSF results were mapped onto the corresponding protein structures, and showed a reduction in the atomic motion of residues within the DNA-binding pocket in the K35G/E198R mutant, suggesting that the structure in this region is more stable (Figure 5A,B). This reduction in flexibility may indicate that the conformation of the active site is more compact or ordered, which could enhance the binding affinity between the protein and DNA.

To explore the correlated motions between UvsX residues and their functional domains, DCCM was calculated and analyzed (Figure 5C,D). The K35G/E198R mutant exhibits a reduction in positive correlations compared to WT, potentially indicating increased independence in residue movements and enhanced flexibility and local degrees of freedom within the structure. This dynamic decoupling may confer greater flexibility during critical reaction steps, allowing the mutant to more effectively facilitate the transition from presynaptic filament formation to homologous recombination.

The P-loop within the ATP binding core structure and the L1 and L2 involved in DNA binding were identified as critical regions. There is a stronger L1-L1 positive correlation between monomer I and monomer II in K35G/E198R compared to WT, which may allow for more efficient coordination of both activities between monomers. The foregoing may result in a greater synergy of K35G/E198R in exercising RPA function.

The free energy landscape (FEL) was used to identify stable conformations and conformational transition paths. Structural stability is measured using the free energy basin, with lower values indicating larger structural population, *i.e.*, better stability. The FEL plot was constructed by combining three MD trajectories (Figure 6). The WT exhibited three distinct deep energy wells, indicating high energy barriers for conformational transitions. In contrast, the K35G/E198R mutant showed two connected wells, suggesting greater flexibility. Overall, the deep energy wells of the K35G/E198R mutant are more spaced compared to the WT, facilitating more efficient conformational transitions, which may enhance the protein's functional adaptability.

The lowest free energy corresponds to the most stable conformation, representing the energetically favored state under equilibrium conditions. The proximity of the deep wells in the K35G/E198R mutant suggests that it can transition to this most stable state more easily. Meanwhile, the lowest energy conformations from each simulation trajectory were selected for comparison. The conformation of WT showed the unraveling of α -helices and unfolding of β -sheets compared to the K35G/E198R. These structural alterations were predominantly concentrated around the L1 and L2 domains, which are required for protein–DNA binding.

3.4. Binding Properties of K35G/E198R to ATP and DNA. As the molecular surface electrostatic potential (MEP) map (Figure 7A–D) showed, the surface potential of the ATP binding pockets of WT and K35G/E198R are equally matched. K35G/E198R exhibits an enlarged ATP binding

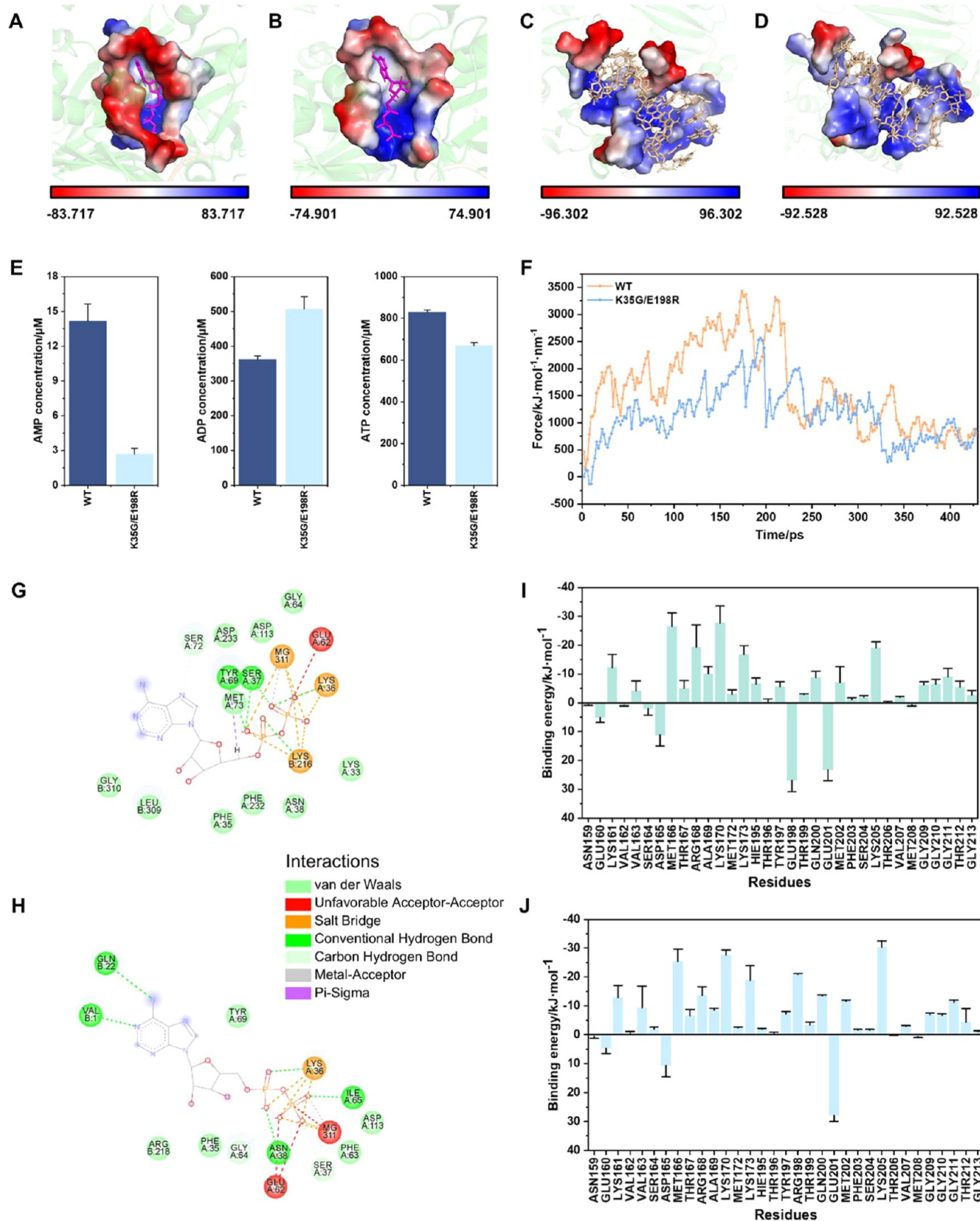


Figure 7. (A–D) MEP map of WT and K35G/E198R. ATP binding pocket for (A) WT and (B) K35G/E198R. DNA binding pocket for (C) WT and (D) K35G/E198R. White regions indicate neutral electrostatic potential, while red and blue correspond to negative and positive potentials, respectively. (E) AMP, ADP, and ATP content of the system after hydrolysis of ATP by WT and K35G/E198R. (F) ADP pull average force. 2D interaction plots of UvsX (G) WT, (H) K35G/E198R and ADP at 100 ps. DNA binding residue energy contributions in UvsX (I) WT, (J) K35G/E198R.

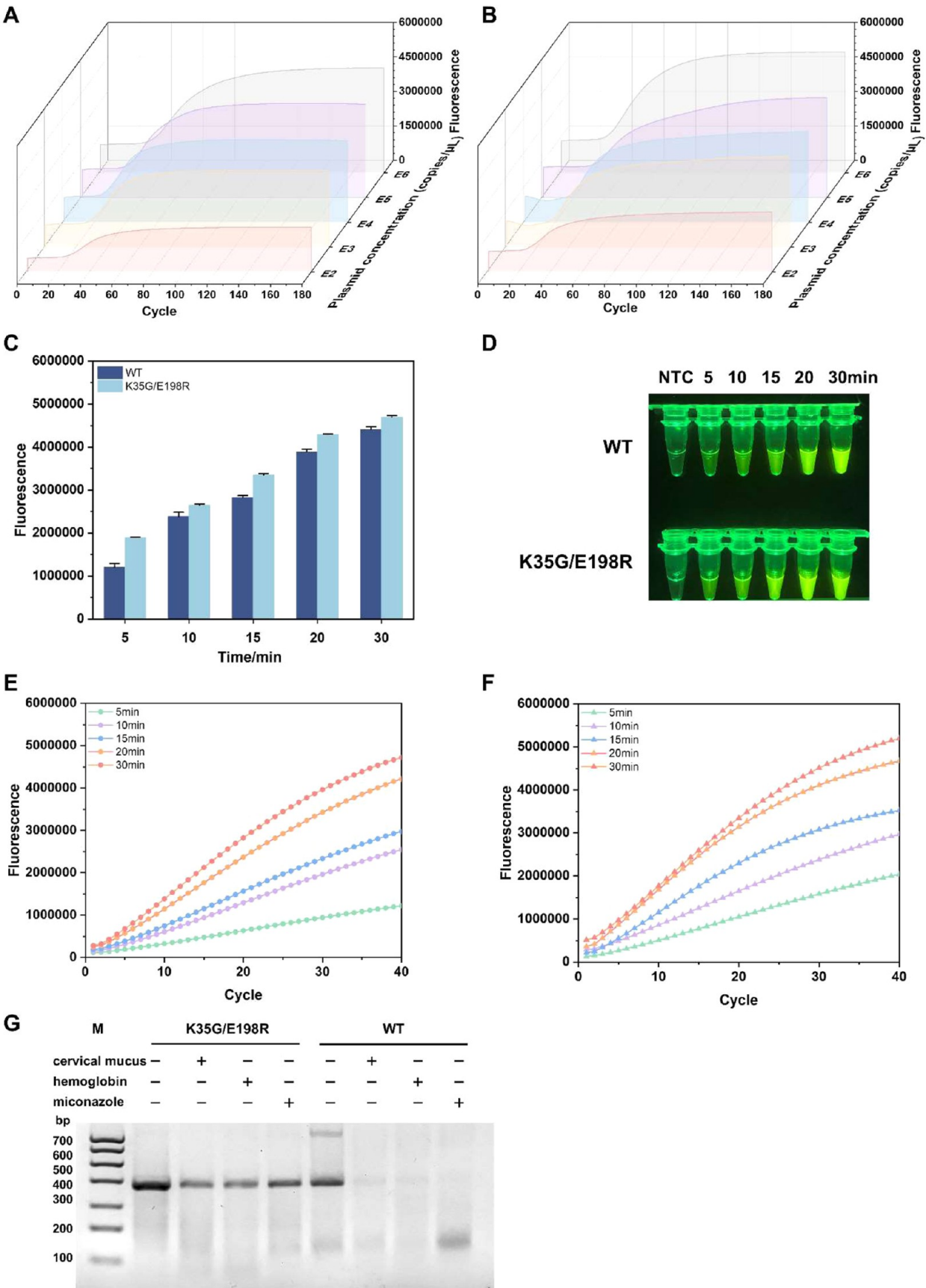


Figure 8. Real-time RPA reaction with different copy number plasmids as templates for sensitivity detection: (A) WT and (B) K35G/E198R. CRISPR/Cas12a assay of WT with K35G/E198R at different RPA reaction times. (C) Fluorescence values. (D) Fluorescence system. And its real-time fluorescence profiles of CRISPR/Cas12a assays for (E) WT and (F) K35G/E198R. (G) Tolerance testing of WT and K35G/E198R to different inhibitors in RPA reaction.

pocket, which allows easier entry and exit of ATP or ADP and reduces AMP accumulation. ATP hydrolysis produces ADP, which can further hydrolyze into AMP in the active pocket, potentially causing toxic accumulation.⁵¹ Experiments verified that K35G/E198R hydrolyzed ATP with significantly less AMP and more ADP generated compared with WT (Figure 7E). Thus, the enlarged ATP-binding pocket in the K35G/E198R mutant enhances the efficient exchange of ATP and its hydrolysis products ADP. This promotes faster catalytic turnover by preventing ADP accumulation and its subsequent conversion to AMP and thus enables rapid energy cycling to accelerate reactions. This effect is supported by kinetic data showing that K35G/E198R has a higher $V_{\max}$ and a lower K_m compared to WT, indicating that it has a higher catalytic efficiency, ATP turnover capacity and ATP binding affinity (Figure S8A,B). Additionally, according to the constant velocity pulling dynamics of ADP with WT and K35G/E198R, WT exhibited higher pulling force (Figure 7F). This is due to the stronger hydrogen bonds, salt bridges and van der Waals interactions between WT and ADP (Figure 7G,H), indicating more stable binding of ADP with the WT structure. Furthermore, the positively charged surface area of the DNA binding pocket in K35G/E198R is larger than WT. By improving electrostatic interactions and raising the affinity for DNA, this alteration probably contributes to more stable and effective DNA binding, confirming the effects of elevated RPA activity.

To evaluate the impact of different residues on UvsX binding properties during interactions with ATP or DNA, we calculated the changes in binding energy for both ATP and DNA (Figure S9). The results indicate that compared to WT, the K35G/E198R mutant reflects enhanced ATP and DNA binding affinity. To precisely catch the beneficial residues involved in DNA recognition and binding, we analyzed the amino acids inside the DNA binding pocket. Their energy contribution during DNA interaction was also evaluated (Figure 7I,J). The residues that interact with DNA are primarily located in L1 and L2. The increased positive charge of amino acid 198 at L2, which changes to arginine, causes an electrostatic contact with DNA, hence improving the stability of the UvsX-DNA complex and its affinity. This also correlates with the increased activity of the experimental results, allowing the MD simulations to confirm the experimental accuracy.

3.5. APPLICATION OF K35G/E198R IN RPA-CRISPR COUPLING FOR HPV16 DETECTION

K35G/E198R also showed several notable performance improvements in addition to having better RPA activity compared to WT. Such as shorter reaction times, higher sensitivity, and better inhibitor tolerance. The HPV16-RPA assay was used to evaluate the performance of K35G/E198R. Real-time amplification curves of WT and K35G/E198R using HPV16 plasmids with different copy numbers as templates are shown in Figure 8A,B. Compared with WT, K35G/E198R exhibits a higher fluorescence signal at the same copy number and can achieve effective amplification at lower concentrations of template DNA, thereby increasing the sensitivity of the assay. RPA-CRISPR/Cas12a coupling was used to detect the products produced by WT and K35G/E198R at different reaction times. The fluorescence values of both WT and K35G/E198R mutant increased with time, whereas significant results were observed at 15 min for K35G/E198R, and the signals observed at 20 min could be almost equal to the signal

values of WT at 30 min (Figure 8C,D). The improved performance of RPA with K35G/E198R plays a crucial role in enabling rapid and highly sensitive nucleic acid amplification and targeted detection, thereby greatly enhancing the efficiency of molecular diagnosis.

The clinical samples used for HPV16 detection are usually cervical exfoliated cells, so in the RPA reaction, cervical mucus, hemoglobin, and miconazole were respectively introduced to evaluate the tolerance of UvsX to inhibitors. K35G/E198R was disturbed to a lesser extent by the inhibitor, whereas WT showed an almost complete absence of RPA activity in the system supplemented with cervical mucus and hemoglobin (Figure 8G). The tests indicated that the method has a wide range of practical applications in the diagnosis of HPV infections and epidemiological investigations. K35G/E198R is expected to make RPA an important tool for HPV detection in the future due to its high sensitivity and rapidity characteristics. This will enable RPA to provide reliable results in resource-limited areas (e.g., remote areas or on-site testing) and mass screening, supporting point-of-care testing (POCT).

4. CONCLUSIONS

This study investigates the role of UvsX recombinase in RPA and explores its directed engineering through virtual saturation and consensus mutagenesis strategies to enhance DNA binding and ATP hydrolysis activities. Notably, several mutants (e.g., E198N, K35G) showed a $39 \pm 6\%$ increase in RPA activity, while the double mutant K35G/E198R showed a significant increase of up to $43 \pm 4\%$. CD and FP analyses confirmed the effects of these mutations on the enzyme's secondary structure and DNA affinity. MD simulations further elucidated how specific mutations affect ATP binding and DNA interaction. HPLC analysis, kinetic data of hydrolyzed ATP and ADP pull average force analysis confirmed that K35G/E198R could reduce the cumulative toxic effects of AMP. Through engineering UvsX recombinase to overcome inherent limitations in ATP utilization and strand exchange fidelity, this study creates new opportunities for its application in CRISPR-integrated POCT. Further modifications to enhance the thermal stability and catalytic activity of UvsX are expected to broaden its applications in diverse recombinase-based technologies, including *in vitro* DNA assembly, synthetic biology, and high-temperature molecular diagnostics.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.biochem.5c00098>.

Primer sequences; RPA amplification sequences; ΔT_m comparison of all proteins, CD spectra and FP results analysis of WT and all positive mutants; additional molecular dynamics simulation data for WT and K35G/E198R and kinetic analysis of hydrolyzed ATP (PDF)

Accession Codes

The structure of UvsX used in this study has PDB Accession Number 3IO5, which corresponds to the UniProt Accession Number P04529.

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Key Research and Development Program of China (2022YFC2106300), the National Natural Science Foundation of China (No. 22278314, 22078239), and Enterprise horizontal entrustment 2023GKF-0146.

ABBREVIATIONS

RPA, recombinase polymerase amplification; ssDNA, single-strand DNA; dsDNA, double-strand DNA; WT, wild type;

CD, circular dichroism; FP, fluorescence polarization; DSF, differential scanning fluorimetry

REFERENCES

- (1) Liu, J.; Ehmsen, K. T.; Heyer, W.; Morrical, S. W. Presynaptic filament dynamics in homologous recombination and DNA repair. *Crit. Rev. Biochem. Mol. Biol.* **2011**, *46*, 240–270.
- (2) Przeworski, G. C. M.; Przeworski, M. An evolutionary view of human recombination. *Nat. Rev. Genet.* **2007**, *8*, 23–24.
- (3) Imkeller, K.; Wardemann, H. Assessing human B cell repertoire diversity and convergence. *Immunol. Rev.* **2018**, *284*, 51–66.
- (4) Kolinjivadi, A. M.; Sannino, V.; de Antoni, A.; Técher, H.; Baldi, G.; Costanzo, V. Moonlighting at replication forks—a new life for homologous recombination proteins BRCA1, BRCA2 and RAD51. *FEBS Lett.* **2017**, *591*, 1083–1100.
- (5) Anand, R.; Beach, A.; Li, K.; Haber, J. Rad51-mediated double-strand break repair and mismatch correction of divergent substrates. *Nature* **2017**, *544*, 377–380.
- (6) Lieber, M. R. The mechanism of double-strand DNA break repair by the nonhomologous DNA end-joining pathway. *Annu. Rev. Biochem.* **2010**, *79*, 181–211.
- (7) Jenkins, S. S.; Mukherjee, S.; Heyer, W. D. DNA repair by homologous recombination. *Encycl. Cell Biol.* **2016**, *1*, 456–467.
- (8) Ameiziane, N.; May, P.; Haitjema, A.; van de Vrugt, H. J.; van Rossum-Fikkert, S. E.; Ristic, D.; Williams, G. J.; Balk, J.; Rockx, D.; Li, H.; Rooimans, M. A.; Oostra, A. B.; Velleuer, E.; Dietrich, R.; Bleijerveld, O. B.; Altelaar, A. F. M.; Meijers-Heijboer, H.; Joenje, H.; Glusman, G.; Roach, J.; Hood, L.; Galas, D.; Wyman, C.; Balling, R.; den Dunnen, J.; de Winter, J. P.; Kanaar, R.; Gelinas, R.; Dorsman, J. C. A novel Fanconi anaemia subtype associated with a dominant-negative mutation in RAD51. *Nat. Commun.* **2015**, *6*, No. 8829.
- (9) Li, X.; Heyer, W. Homologous recombination in DNA repair and DNA damage tolerance. *Cell Res.* **2008**, *18*, 99–113.
- (10) Juma, K. M.; Takita, T.; Ito, K.; Yamagata, M.; Akagi, S.; Arikawa, E.; Kojima, K.; Biyani, M.; Fujiwara, S.; Nakura, Y.; Yanagihara, I.; Yasukawa, K. Optimization of reaction condition of recombinase polymerase amplification to detect SARS-CoV-2 DNA and RNA using a statistical method. *Biochem. Biophys. Res. Commun.* **2021**, *567*, 195–200.
- (11) Wang, Y.; Chen, R.; Nie, X.; Zhong, Z.; Li, C.; Li, K.; Huang, W.; Fu, X.; Liu, J.; Nie, B. Rapid and sensitive detection of potato virus Y by isothermal reverse transcription-recombinase polymerase amplification assay in potato. *Mol. Cell. Probes* **2020**, *50*, No. 101505.
- (12) Liu, H.; Wang, J.; Zeng, H.; Liu, X.; Jiang, W.; Wang, Y.; Ouyang, W.; Tang, X. RPA-Cas12a-FS: A frontline nucleic acid rapid detection system for food safety based on CRISPR-Cas12a combined with recombinase polymerase amplification. *Food Chem.* **2021**, *334*, No. 127608.
- (13) Wang, H.; Li, Z.; Ren, H.; Zhang, C.; Xiao, D.; Li, Y.; Hou, X.; Liu, T. Regulatory interaction of BcWRKY33A and BcHSA4A promotes salt tolerance in non-heading Chinese cabbage. *Hortic. Res.* **2022**, *9*, No. uhac113.
- (14) Lobato, I. M.; O'Sullivan, C. K. Recombinase polymerase amplification: basics, applications and recent advances. *TrAC, Trends Anal. Chem.* **2018**, *98*, 19–35.
- (15) Lillis, L.; Lehman, D.; Singhal, M. C.; Cantera, J.; Singleton, J.; Labarre, P.; Toyama, A.; Piepenburg, O.; Parker, M.; Wood, R.; Overbaugh, J.; Boyle, D. S. Non-instrumented incubation of a recombinase polymerase amplification assay for the rapid and sensitive detection of proviral HIV-1 DNA. *PLoS One* **2014**, *9*, No. e108189.
- (16) Xu, C.; Li, L.; Jin, W.; Wan, Y. Recombinase polymerase amplification (RPA) of CaMV-35S promoter and nos terminator for rapid detection of genetically modified crops. *Int. J. Mol. Sci.* **2014**, *15*, 18197–18205.
- (17) Li, J.; Macdonald, J.; Von Stetten, F. Review: a comprehensive summary of a decade development of the recombinase polymerase amplification. *Analyst* **2019**, *144*, 31–67.

- (18) Maher, R. L.; Morrical, S. W. Coordinated binding of single-stranded and double-stranded DNA by UvsX recombinase. *PLoS One* **2013**, *8*, No. e66654.
- (19) Pan, Y.; Zhao, C.; Fu, W.; Yang, S.; Lv, S. Comparative analysis of structural dynamics and allosteric mechanisms of RecA/Rad51 family proteins: Integrated atomistic MD simulation and network-based analysis. *Int. J. Mol. Sci.* **2024**, *261*, No. 129843.
- (20) Gajewski, S.; Webb, M. R.; Galkin, V.; Egelman, E. H.; Kreuzer, K. N.; White, S. W. Crystal Structure of the Phage T4 Recombinase UvsX and Its Functional Interaction with the T4 SF2 Helicase UvsW. *J. Mol. Biol.* **2011**, *405*, 65–76.
- (21) van der Heijden, T.; Modesti, M.; Hage, S.; Kanaar, R.; Wyman, C.; Dekker, C. Homologous Recombination in Real Time: DNA Strand Exchange by RecA. *Mol. Cell* **2008**, *30*, 530–538.
- (22) Collier, R.; Thomson, J. G.; Thilmony, R. A versatile and robust Agrobacterium-based gene stacking system generates high-quality transgenic Arabidopsis plants. *Plant J.* **2018**, *95*, 573–583.
- (23) Wu, J.; Wang, M.; Yang, X.; Yi, C.; Jiang, J.; Yu, Y.; Ye, H. A non-invasive far-red light-induced split-Cre recombinase system for controllable genome engineering in mice. *Nat. Commun.* **2020**, *11*, No. 3708.
- (24) Cautereels, C.; Smets, J.; Bircham, P.; De Ruysscher, D.; Zimmermann, A.; De Rijk, P.; Steensels, J.; Gorkovskiy, A.; Masschelein, J.; Verstrepen, K. J. Combinatorial optimization of gene expression through recombinase-mediated promoter and terminator shuffling in yeast. *Nat. Commun.* **2024**, *15*, No. 1112.
- (25) Shilling, P. J.; Mirzadeh, K.; Cumming, A. J.; Widesheim, M.; Köck, Z.; Daley, D. O. Improved designs for pET expression plasmids increase protein production yield in *Escherichia coli*. *Commun. Biol.* **2020**, *3*, 214–222.
- (26) Abramson, J.; Adler, J.; Dunger, J.; Evans, R.; Green, T.; Pritzel, A.; Ronneberger, O.; Willmore, L.; Ballard, A. J.; Bambrick, J.; Bodenstein, S. W.; Evans, D. A.; Hung, C.; O'Neill, M.; Reiman, D.; Tunyasuvunakool, K.; Wu, Z.; Žemgulytė, A.; Arvaniti, E.; Beattie, C.; Bertolli, O.; Bridgland, A.; Cherepanov, A.; Congreve, M.; Cowen-Rivers, A. I.; Cowie, A.; Figurnov, M.; Fuchs, F. B.; Gladman, H.; Jain, R.; Khan, Y. A.; Low, C. M. R.; Perlin, K.; Potapenko, A.; Savy, P.; Singh, S.; Stecula, A.; Thillaisundaram, A.; Tong, C.; Yakneen, S.; Zhong, E. D.; Zielinski, M.; Židek, A.; Bapst, V.; Kohli, P.; Jaderberg, M.; Hassabis, D.; Jumper, J. M. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **2024**, *630*, 493–500.
- (27) Wang, E.; Sun, H.; Wang, J.; Wang, Z.; Liu, H.; Zhang, J. Z. H.; Hou, T. End-point binding free energy calculation with MM/PBSA and MM/GBSA: strategies and applications in drug design. *Chem. Rev.* **2019**, *119*, 9478–9508.
- (28) Schymkowitz, J.; Borg, J.; Stricher, F.; Nys, R.; Rousseau, F.; Serrano, L. The FoldX web server: an online force field. *Nucleic Acids Res.* **2005**, *33*, W382–W388.
- (29) Martin-Malpartida, P.; Hausvik, E.; Underhaug, J.; Torner, C.; Martinez, A.; Macias, M. J. HTSDSF Explorer, a novel tool to analyze high-throughput DSF screenings. *J. Mol. Biol.* **2022**, *434*, No. 167372.
- (30) Li, X.; Zhang, C. Using differential scanning fluorimetry (DSF) to detect ligand binding with purified protein. *Methods Mol. Biol.* **2021**, *2213*, 183–186.
- (31) Schneider, C. A.; Rasband, W. S.; Kevin, W. S. R. NIH Image to ImageJ: 25 years of image analysis. *Nat. Methods* **2012**, *9*, 671–675.
- (32) Worsfold, P.; McKelvie, I.; Monbet, P. Determination of phosphorus in natural waters: A historical review. *Anal. Chim. Acta* **2016**, *918*, 8–20.
- (33) Race, A. M.; Palmer, A. D.; Dexter, A.; Steven, R. T.; Styles, I. B.; Bunch, J. SpectralAnalysis: software for the masses. *Anal. Chem.* **2016**, *88*, 9451–9458.
- (34) Pandiscia, L. JWMVS-529 CD Multivariate SSE Analysis Program. <https://jascoinc.com/applications/jwmvs-529-cd-multivariate-sse-analysis-program/> (accessed October 20, 2024).
- (35) Pandiscia, L. Multivariate Secondary Structure Estimation (SSE) with High Throughput CD. <https://jascoinc.com/applications/high-throughput-cd-multivariate-sse/> (accessed October 20, 2024).
- (36) Zhang, R.; Zhang, L.; Wu, L.; You, S.; Su, R.; Qi, W. Rapid detection of HPV16 utilizing recombinase polymerase amplification with the employment of an extremely low concentration of the probe. *Anal. Methods* **2024**, *16*, 7557–7566.
- (37) Lu, T.; Chen, F. Multiwfn: a multifunctional wavefunction analyzer. *J. Comput. Chem.* **2012**, *33*, 580–592.
- (38) Lu, T. A comprehensive electron wavefunction analysis toolbox for chemists, Multiwfn. *J. Chem. Phys.* **2024**, *161*, No. 82503.
- (39) Zhang, J.; Lu, T. Efficient evaluation of electrostatic potential with computerized optimized code. *Phys. Chem. Chem. Phys.* **2021**, *23*, 20323–20328.
- (40) Lu, T. Sobtop, Version 1.0(dev5). <http://sobereva.com/soft/Sobtop> (accessed December 4, 2024).
- (41) Abraham, M. J.; Murtola, T.; Schulz, R.; Páll, S.; Smith, J. C.; Hess, B.; Lindahl, E. GROMACS: high performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* **2015**, *1–2*, 19–25.
- (42) Lindorff-Larsen, K.; Piana, S.; Palmo, K.; Maragakis, P.; Klepeis, J. L.; Dror, R. O.; Shaw, D. E. Improved side-chain torsion potentials for the Amber ff99SB protein force field. *Proteins: Struct., Funct., Bioinf.* **2010**, *78*, 1950–1958.
- (43) Grant, B. J.; Rodrigues, A. P. C.; ElSawy, K. M.; McCammon, J. A.; Caves, L. S. D. Bio3d: an R package for the comparative analysis of protein structures. *Bioinformatics* **2006**, *22*, 2695–2696.
- (44) Chen, J.; Zeng, Q.; Wang, W.; Sun, H.; Hu, G. Decoding the identification mechanism of an SAM-III riboswitch on ligands through multiple independent Gaussian-accelerated molecular dynamics simulations. *J. Chem. Inf. Model.* **2022**, *62*, 6118–6132.
- (45) Berendsen, H. J. C.; Postma, J. P. M.; van Gunsteren, W. F.; DiNola, A.; Haak, J. R. Molecular dynamics with coupling to an external bath. *J. Chem. Phys.* **1984**, *81*, 3684–3690.
- (46) Essmann, U.; Perera, L.; Berkowitz, M. L.; Darden, T.; Lee, H.; Pedersen, L. G. A smooth particle mesh Ewald method. *J. Chem. Phys.* **1995**, *103*, 8577–8593.
- (47) Humphrey, W.; Dalke, A.; Schulten, K. VMD: visual molecular dynamics. *J. Mol. Graphics* **1996**, *14*, 33–38.
- (48) Zhang, J. s_mmpbsa, Version 0.6.5. https://github.com/supernova4869/s_mmpbsa (accessed December 4, 2024).
- (49) Islam, R.; Parves, M. R.; Paul, A. S.; Uddin, N.; Rahman, M. S.; Mamun, A. A.; Hossain, M. N.; Ali, M. A.; Halim, M. A. A molecular modeling approach to identify effective antiviral phytochemicals against the main protease of SARS-CoV-2. *J. Biomol. Struct. Dyn.* **2021**, *39*, 3213–3224.
- (50) Pourmand, S.; Zareei, S.; Shahlaei, M.; Moradi, S. Inhibition of SARS-CoV-2 pathogenesis by potent peptides designed by the mutation of ACE2 binding region. *Comput. Biol. Med.* **2022**, *146*, No. 105625.
- (51) Villanueva-Paz, M.; Cotán, D.; Garrido-Maraver, J.; Oropesa-Ávila, M.; de la Mata, M.; Delgado-Pavón, A.; de Laveria, I.; Alcocer-Gómez, E.; Álvarez-Córdoba, M.; Sánchez-Alcázar, J. AAMPK Regulation of Cell Growth, Apoptosis, Autophagy, and Bioenergetics. In *AMP-activated Protein Kinase*, *Experientia Supplementum*; Springer, 2016; Vol. 107, pp 45–71.